

CostVision

AI-Powered Should-Cost Intelligence for Modern Engineering

Designed & Developed by Avinash Bhosale

Cost Engineering & Digital Innovation

21 Commodity Models

AI Agent (NLP)

CAD-to-Cost AI

DFM / DFA Layer

20 Regions · 10 FX

Assembly BOM Rollup

Wright's Law Learning Curve

Supplier Quote Comparison

Cloud Team Sync

THE CHALLENGE

Why CostVision? Traditional Costing is Broken

Manual effort, weeks of delay, and supplier dependency cost organisations millions annually.

 Weeks, Not Minutes

Manual should-cost takes 2-4 weeks per part. Sourcing decisions stall. Programme timelines slip.

 Supplier Dependency

Without independent cost intelligence, engineers accept quotes at face value — no leverage, no transparency.

 Inconsistent Methodology

Every engineer calculates cost differently. No standard model. No auditable assumptions. No cross-team comparability.

 No Regional Visibility

Labour, machine, material and energy rates vary dramatically by country. Manual adjustment is error-prone.

 Late-Stage Cost Surprises

Cost issues discovered at SOP — not at concept. DFM problems missed. Expensive redesigns. Margin erosion at launch.

 No Supplier Quote Benchmark

Buyers lack a defensible floor price for negotiation. Supplier margins are opaque and unchallenged.

80
%

of part cost is locked in at the design stage — yet most cost analysis happens after design freeze. CostVision shifts cost intelligence to where it matters: concept phase.

THE SOLUTION

What is CostVision?

A world-class should-cost engine with AI at every layer — from natural language input to DFM/DFA output.



AI Agent (Natural Language)

Describe a part in plain English. AI auto-builds the complete cost model: material, routing, machines, operations.



CAD-to-Cost Automation

Upload STEP / IGES / photo. AI extracts geometry, infers material, selects process model — full cost in minutes.



21 Engineering-Grade Cost Models

Physics-based models for every major commodity. Routing logic, yield curves, tooling amortisation — all encoded.



Assembly BOM Rollup

Cost entire assemblies — not just single parts. Multi-level BOM with per-component commodity models.



Learning Curve / Wright's Law

Project cost at any volume milestone. Configure learning curve %. Essential for LTA negotiations.



Supplier Quote Comparison

Log supplier quotes. Compare vs should-cost instantly. Identify margin gaps. Generate negotiation talking points.



DFM / DFA Intelligence

Bolt-on AI scores manufacturability 1–10. Critical issues, saving potential, actionable recommendations.



20 Regions · 10 Currencies

Real-time labour, machine, material and energy rates. Currency auto-switches when region is selected.



Export, Scenarios & Team Sync

6-sheet Excel, PDF report, A/B/C scenario comparison, tornado sensitivity chart, cloud team sync.

GAME-CHANGING CAPABILITY

AI Agent — Natural Language Should-Costing

The fastest path from part concept to full cost breakdown: just describe the part in plain English.

How It Works

1. Describe

Type a part description: 'Aluminium bracket, 6082-T6, approximately 200×100×50mm, 3 machined holes, anodised finish, 5,000/year UK production'

2. AI Analyses

Claude AI interprets the description, infers geometry, selects the most cost-effective manufacturing route, and chooses appropriate machines and labour grades

3. Auto-Populate

The cost model is automatically populated: material weight, 3 machining operations, cycle times, OEE, tooling NRE — all filled without manual input

4. Full Result

Click Calculate. Full 8-bucket cost breakdown, AI insights, DFM/DFA score, sensitivity analysis, and export — all ready in under 60 seconds

● CostVision AI Agent

You: Describe a 4-layer FR4 PCB, 100×80mm, ENIG finish, flying probe test, 10,000/yr, UK manufacture

AI: Got it. Analysing your PCB specification...

- 4-layer FR4 standard, 100×80 mm board
- ENIG surface finish (+22% vs HASL)
- Flying probe electrical test
- UK pricing region applied
- NRE amortised over 10,000 boards

You: What is the should-cost?

AI: Should-Cost Result: £ 4.18 per board

- PCB Fab (material + process): £ 3.24
- Test (flying probe): £ 0.71
- NRE amortised: £ 0.23

DFM Score: 8.5/10 — Good manufacturability

Key Benefits

AUTOMATION

CAD-to-Cost: Geometry to Should-Cost Automatically

Upload a STEP file or part photo — AI extracts features, infers material, and generates a full cost model.

Supported formats: .STEP .IGES .STP .IGS .JPG .PNG .HEIC (drag-drop or file picker)

What AI Extracts from CAD / Photo

- Part geometry: envelope dimensions, volume, wall thickness, surface area
- Feature recognition: holes, pockets, ribs, undercuts, threads, datum features
- Material inference: visual texture & density → grade selection (e.g. AI 6082-T6)
- Process selection: geometry → most cost-effective manufacturing route
- Cycle time: material removal rate, tool path length, fixture & setup strategy
- Complexity: Simple / Moderate / Complex → machine type & OEE selection
- DFM pre-check: thin walls, tight radii, undercuts flagged before costing begins

AI Analysis Result — Aluminium Bracket

AI Detected Material

AI 6082-T6

Suggested Process

3-Axis CNC Milling + Drilling

Envelope (L×W×H)

220 × 140 × 85 mm

Est. Net Weight

0.82 kg

Complexity

Moderate — 3 setups

DFM Pre-check

1 flag: 3 setups → consolidate

Should-Cost Preview

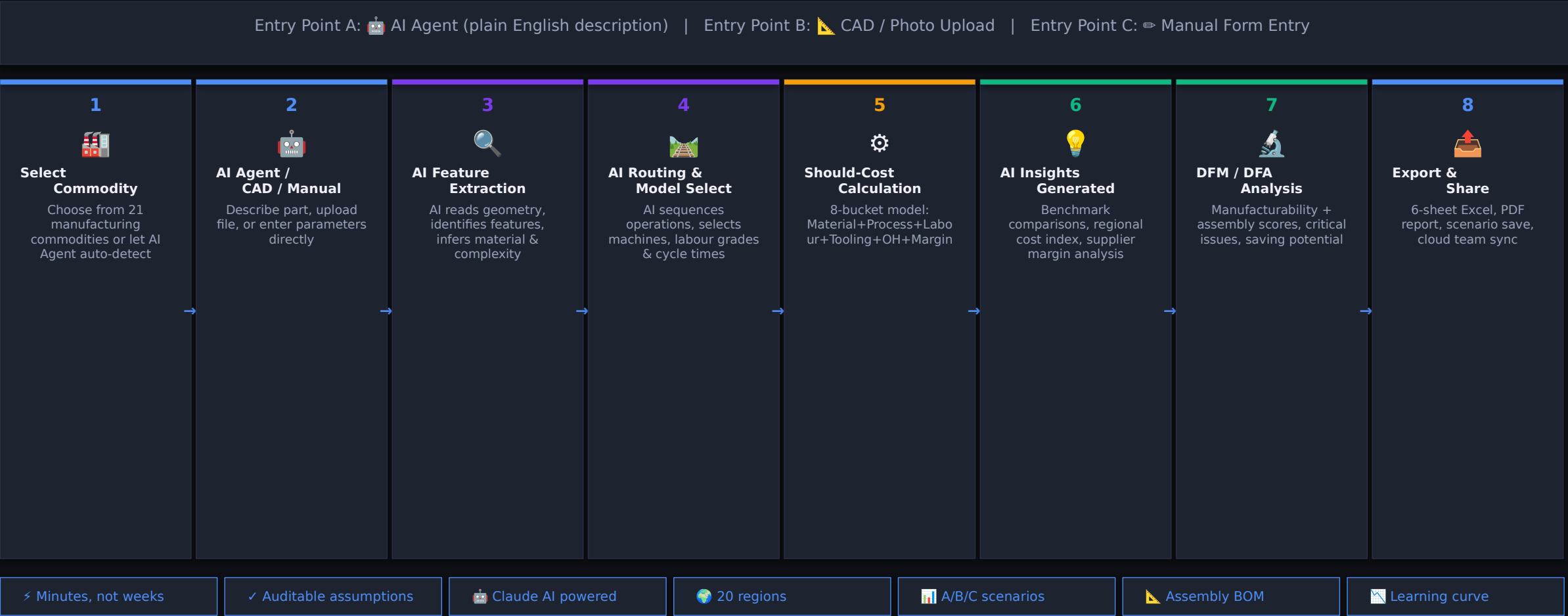
£ 28.40



WORKFLOW

AI Agent — 8-Step End-to-End Flow

Three entry points. One intelligent engine. Full cost intelligence in under 5 minutes.



FULL COVERAGE

21 Manufacturing Commodities — One Platform

Every major manufacturing process — all modelled to engineering depth with full routing, yield, and tooling logic.

 CNC Machining 3/4/5-axis, turning, grinding	 Casting HPDC, gravity, investment, sand	 Cast + Machine Casting + post-ops + HT	 Forging Hot, warm, cold die forging	 Sheet Metal Progressive/transfer die	 Sheet Metal Fab Laser · punch · press brake · MIG	 Injection Moulding Thermoplastics, multi-cavity
 Blow Moulding HDPE, PET, extrusion blow	 Extrusion Profile, pipe, sheet, rod	 Thermoforming Vacuum, pressure, twin-sheet	 Rotational Moulding Large hollow shapes, LLDPE	 Rubber Moulding Compression, injection, LSR	 Composites CFRP/GFRP, autoclave, RTM, AFP	 Painting / Coating E-coat, powder, wet, anodise
 BIW Assembly Spot weld, MIG, hem, bond	 PCB Fabrication 2-16L, HDI, rigid-flex, RF/μwave	 PCBA / SMD SMT, TH, reflow, conformal coat	 Wiring Harness Cut-strip-crimp, test, sub-asm	 Assembly BOM Rollup Multi-part BOM, full cost rollup	 Extrusion (Profile) Continuous linear profiles	 AI Agent Natural language → any commodity

ENGINEERING DEPTH

Should-Cost Model Architecture

Built on aPriori-calibrated benchmarks, IPC standards, and first-principles manufacturing engineering.

 Material Database

30+ materials: alloy steels, aluminium grades, engineering plastics, elastomers, composites, PCB laminates, copper alloys. FX-adjusted per region.

 Machine & Rate Library

50+ machine types: CNC centres, presses, moulding machines, SMT lines, reflow ovens, test equipment. Hourly rates editable per project and region.

 Routing & Process Logic

AI-driven operation sequencing. Each operation: machine ID, cycle time (hr), OEE, setup time, labour grade, overhead allocation.

 Tooling

Dies, mould volume with electrodes.

 Yield & Scrap Models

First-pass yield from 10 penalty factors: layer count, trace width, via type, HDI structure, impedance control, BGA pitch, alloy grade, section thickness.

 Regional Cost Multipliers

20 regions — each with calibrated labour tiers (skilled/semi-skilled/engineer/inspector), energy costs, and machine hour rates.

 Benchmarks & Sensitivity

Every cost element benchmarked vs aPriori/Cleansheet ranges. Tornado chart shows top 10 cost drivers at $\pm 10\%$. Scenario A/B/C comparison.

 Learning

Configure 8: volume-driven pricing and

 Supplier Quote Comparison

Log supplier quotes (name, date, price, currency). Compare vs should-cost instantly — identify margin gaps and overhead inflation.

 AI-Driven Assumptions

Where geometry is unknown, Claude AI fills assumptions transparently — material, complexity, process, geometry — with explanation of every decision.

ADVANCED CAPABILITIES

Advanced Cost Modelling & Intelligence Features

CostVision goes beyond basic costing — it models commercial reality with precision.

Wright's Law Learning Curve

Configure any learning curve % (default 85%). As volume doubles, cost reduces by the curve %.

- Model programme cost at any volume milestone
- Critical for LTA negotiations and make-vs-buy decisions
- Visualise break-even volume vs manual costing cost

Assembly BOM Rollup

Build multi-level assemblies from any combination of 21 commodities:

- Add components one-by-one with commodity type, weight, volume
- Each component costed using its own should-cost model
- Roll up to a total assembly cost with full per-component breakdown
- Example: BIW + Painting + Harness + PCBA = full module should-cost

Supplier Quote Comparison

- Enter supplier name, quote date, price, currency, and FX rate
- CostVision converts to GBP and compares vs should-cost
- Calculates margin gap: 'Supplier is 18% above should-cost'
- Generates data-driven negotiation talking points
- Track multiple suppliers per part for competitive benchmarking

Sensitivity Analysis (Tornado Chart)

- $\pm 10\%$ variation applied to each cost driver independently
- Top 10 drivers ranked by impact (tornado chart)
- Identify which inputs most need validation before negotiation
- Export the chart to include in supplier presentations

Scenario A/B/C Comparison

- Save any configuration as a named scenario
- Compare baseline vs target vs stretch cost side-by-side
- Delta view shows where costs increased or decreased
- Export all scenarios to JSON for team sharing

AI INTELLIGENCE LAYER

DFM / DFA — Design for Manufacture & Assembly

A bolt-on AI intelligence layer that runs AFTER the should-cost, without modifying any calculation logic.

Manufacturability Score (DFM)	Assembly Efficiency (DFA)	Total Potential Saving	
8.5 / 10	7.0 / 10	~19.3%	All scores are 1-10 (10 = perfect) Savings are RSS-combined from top 3 issues
DFM Issues — What It Finds Critical: Low material utilisation (<60%), OEE <70%, die cost >18% Major: Below-benchmark OEE (70-80%), high operation count, process cost dominant Minor: Elevated tooling %, overhead burden >18%, supplier margin >18% Opportunity: Near-net-shape conversion, volume increase, multi-cavity tooling	DFA Issues — Assembly Intelligence Automation feasibility: labour-dominated assemblies flagged for robot/cobot study Operation count: multiple setups imply fixturing cost and dimensional variation risk OEE pacing: low OEE in assembly indicates manual pacing or changeover inefficiency Fastener standardisation: variety in fastener drives assembly time and tool changes		
Cost Optimisation Levers (6-8 per part) Automate High-Labour Operations — typically 5-20% saving Near-Net-Shape Material Improvement — 5-15% material cost saving OEE Improvement (TPM programme) — 5-12% machine cost reduction Volume Increase to Dilute Tooling NRE — Quick Win, Low Risk Overhead Rate Negotiation via Open-Book Costing — Quick Win Competitive RFQ to Reduce Supplier Margin — Quick Win	AI Deep Analysis (Claude AI) Click 'Run AI Deep Analysis' after rule-based results are shown Claude provides: Root cause commentary, Priority top-3 actions Supplier negotiation strategy with specific talking points Alternative process recommendations (e.g. casting → HPDC) Risk assessment: red flags in cost structure needing investigation		

GLOBAL COVERAGE

20 Manufacturing Regions — 10 Currencies

Select a region and every rate adjusts automatically — labour, machines, energy, material and currency.

Key Manufacturing Regions (sample)

Region	Currency	Sample Labour Rate	Position
 UK	GBP £	Skilled £26/hr	Industry / Aerospace baseline
 Germany	EUR €	Skilled €32/hr	Premium, high automation
 Poland	EUR €	Skilled €11/hr	Low-cost EU option
 Romania	EUR €	Skilled €7/hr	Ultra-low EU labour
 China	CNY ¥	Skilled ¥28/hr	Global volume manufacturing
 India	INR ₹	Skilled ₹320/hr	Engineering & electronics hub
 Mexico	MXN \$	Skilled MXN82/hr	Near-shore for North America
 USA	USD \$	Skilled \$32/hr	Premium North American rates
 Thailand	THB ฿	Skilled ฿180/hr	Electronics & automotive
 Vietnam	VND ₫	Skilled ₫52k/hr	Lowest-cost electronics hub
 Brazil	BRL R\$	Skilled R\$22/hr	South America manufacturing
 S. Korea	KRW ₩	Skilled ₩22k/hr	Electronics & automotive

Auto-Currency Switch

When you select a manufacturing region (e.g. China), the display currency automatically switches to CNY. No manual FX conversion needed.

Full Rate Adjustment

Every rate adjusts: skilled/semi-skilled/engineer/inspector labour, electricity, gas, machine rates, material prices — all recalculated for the selected region.

Editable Rate Library

All rates are editable. Override any rate with project-specific data, supplier quotes, or latest market prices. Resets to calibrated baseline at any time.

Regional Comparison

AI Insights tab shows estimated cost in all 20 regions simultaneously — instant make-vs-buy regional arbitrage analysis for any part.

THE CONCEPT · EXPLAINED

Agentic vs Autonomous Agentic AI — With Examples

Two levels of "agentic": on the left the AI acts because you asked; on the right it acts on its own, unattended.

Agentic AI

Assisted — you trigger it and stay in the loop; a human approves

AI Agent

Describe a part in plain English — it builds the full cost model for you.

CAD / photo → cost

Upload a STEP file — it measures geometry and infers material & process.

Negotiation coach

Open a part — it drafts the buyer's counter-argument and target price.

Rate-data assistant

Ask a costing question — it answers from your own rates, with citations.

Autonomous Agentic AI

Self-directed — runs unattended on a schedule, within limits you set

Savings monitor

Runs on the server, compares paid vs should-cost, opens findings itself — £0.5M/yr unattended.

Self-audit

Re-checks every estimate for known errors and corrects within bounds — unasked.

Calibration & drift

Learns from logged quotes, re-derives factors, watches for drift continuously.

Outcome-weighted ranking

Learns which findings actually convert and re-prioritises its own queue.

The common thread: you set the boundaries and every action stays glass-box and auditable — autonomy never means the AI sets a price in secret.

NEW IN 2026 · AGENTIC INTELLIGENCE

The Tool Now Checks Its Own Work — and Learns

A deterministic self-audit re-checks every estimate, calibration learns from real quotes, and it is proven on real parts.

Self-Audit — it checks its own homework

A deterministic layer re-runs every estimate looking for known mistakes: a part classed as the wrong process, machining time over the physical finish envelope, a wall thinner than the geometry allows, weight that disagrees with the CAD. It applies bounded corrections and shows its working — the AI never overrules the measured geometry.

Calibration & Drift — learns from actuals

Bulk-import real supplier quotes as a CSV. The engine learns per-segment correction factors, carries a conformal confidence band, and watches for drift — so the estimate tightens as your own history grows. Calibration status now rides on every single estimate, not a hidden report.

Universal Machine-Sizing — right press, any commodity

One dispatcher now sizes the machine for the part across commodities — forging presses, stamping presses, HPDC and moulding tonnage, extrusion and more — instead of a single hard-coded tier. Tooling amortises over annual volume everywhere by one universal rule.

Proven on five real automotive CAD parts

Each was mis-costed on the first pass; the self-audit caught it and corrected it.

Real Part	AI First Pass	What Was Wrong	Corrected
Fuel tank	£216.97	Treated as cast	£24.62 blow-mould
Front bumper	£29.18	Machined, not moulded	£7.79 injection
Servo horn	£333.00	Bulk, not net-shape	£6.40 net-shape
Stub axle	mis-class	Wrong process/weight	Forged steel 8.1 kg
Seat cross-member	engine crash	Sizing gap	£1.11 runs clean

Built to be trusted, not just believed

- 1,005 automated tests across 86 suites — a logic regression fails the build, not the demo
- The self-audit re-checks every estimate before you ever see the number
- The CAD geometry engine ships inside a container whose build is verified in CI

LIVE DEMO DATA

Real-World Cost Benchmarks — Luxury SUV Programme

All figures generated by CostVision's should-cost engine. UK manufacturing region, GBP.

Commodity	Demo Part	Key Inputs	Should-Cost	AI Top Insight	DFM Flag
⚙️ Machining	AI Gearbox Bracket	Al 6082-T6 · 0.85 kg · 3 setups	£ 28.40	Process 44% → near-net-shape saves £8	Major: 3 setups → 5-axis
🏠 HPDC Casting	Transmission Housing	Al A380 · 3.2 kg · 50k/yr	£ 18.70	Die 9% OK. Margin 21% → negotiate	Good — no critical issues
🔥 Sheet Metal	Rear Door Inner	DC04 1.2mm · 4.1 kg · 60k/yr	£ 22.10	Blank nesting 64% → improve saves £2.80	Critical: nesting <65%
💧 Injection	PP Bumper Bracket	PP-TD20 · 0.42 kg · 100k/yr	£ 3.85	Mould 8% — good at this volume	Good — runner waste 15%
🧱 Rubber	EPDM Door Seal	Extrusion · 2.8m · 85 Shore A	£ 4.10	Process 42% — cure cycle dominant	Major: cure time optimise
💻 PCB Fab	4-Layer FR4 ECU Board	100×80mm · ENIG · Flying probe	£ 4.18	NRE 6% per board — increase batch	Good — standard technology
🔧 PCBA	ECU PCBA 85 comp.	SMT + TH · 4 TH joints	£ 32.60	Labour 38% — convert TH to SMT saves £4	Major: high TH labour
🔌 Wiring Harness	Door Module Harness	24 circuits · 2.1m avg	£ 11.80	Labour 58% — automation saves 20%	Critical: labour >50%
🦊 Composites	CFRP Floor Pan	Prepreg CFRP · 1.8 kg · autoclave	£ 185.00	Labour 44% — evaluate ATL/AFP	Major: manual layup cost
🔨 Forging	Steel Control Arm	42CrMo4 · 2.1 kg · 80k/yr	£ 9.20	Die 11% — acceptable at volume	Good — closed-die efficient

REPORTING & COLLABORATION

Export, Share & Collaborate — Enterprise Ready

From individual engineer to global team — CostVision scales to any organisation size.

6-Sheet Excel Export

- Sheet 1: Summary (total cost, breakdown %, region, commodity)
- Sheet 2: Material (alloy, weight, utilisation, price)
- Sheet 3: Operations (machine, labour, cycle time, OEE per step)
- Sheet 4: Machine Rates (rate library snapshot)
- Sheet 5: Labour Rates (regional labour tiers)
- Sheet 6: Traceability (assumptions, AI notes, version, date)

Professional PDF Report

- Section 1: Cover page with part photo & programme details
- Section 2: Cost summary and 8-bucket breakdown chart
- Section 3: Detailed operation-by-operation cost build
- Section 4: Machine and labour rate build-up table
- Section 5: AI insights and saving opportunities
- Section 6: DFM/DFA scores, issues and recommendations

Cloud Team Sync

- Scenarios saved to cloud database (SQLite / PostgreSQL)
- All team members see the same cost baseline
- Secure JWT authentication + OTP email verification
- Full multi-user access — no spreadsheet version conflicts
- Import/export scenarios as JSON for offline sharing

Part Photo & Visual Docs

- Upload part photo (JPG / PNG / HEIC) with any calculation
- Photo appears on the PDF report cover page
- AI Agent accepts photos as input for visual cost estimation
- Drag-and-drop or file picker supported

Enterprise Authentication

- JWT token-based authentication with 24hr expiry
- OTP email verification for account security
- Password reset flow with secure email delivery
- Rate limiting: 10 sign-in attempts / 15 min window

Help Centre & Support

- 7-tab Help Centre: Getting Started, AI Agent, CAD & Photo, Commodities, FAQ (6 items), Glossary (12 terms), Troubleshooting (5 common issues)
- Contact support form built-in to the application

VALUE DELIVERED

Business Impact & Measurable ROI

Quantified improvements across speed, accuracy, and cost reduction that directly impact the bottom line.

70-90%

Reduction in costing time

±8%

Typical cost model accuracy

15-25%

Supplier price reduction via should-cost intel

3×

Faster sourcing decisions

£15M

Saving on £500M spend at 3% improvement

⚡ **Speed — 10× Faster**

What took 2-4 weeks now takes 5-10 minutes. Engineers spend time on decisions, not spreadsheets. Faster RFQ responses, faster programme decisions.

🎯 **Accuracy — Not Estimates**

Physics-based models calibrated to actual shop floor data. ±8% typical accuracy vs actual. Dramatically better than rule-of-thumb or parametric estimates.

💰 **Negotiation Power**

Should-cost models give buyers a defensible floor price for every part. Supplier margins become visible and challengeable with engineering data, not opinion.

🔺 **DFM at Concept Stage**

80% of cost is locked at design. DFM/DFA analysis at concept prevents expensive late-stage redesigns. Engineering issues caught when change is free.

🌐 **Regional Arbitrage**

Instantly compare cost across 20 regions. Identify LCC sourcing opportunities. Quantify make-vs-buy with real data — not assumptions.

📊 **Standardised Methodology**

One consistent cost model across all teams, programmes, and regions. Full audit trail for every assumption. Leadership gets a single source of truth.

📈 **Volume Optimisation**

Learning curve modelling shows cost reduction with volume. Identify the volume at which a new process becomes cost-competitive.

👥 **Team Collaboration**

Cloud sync means all engineers share the same cost baseline. No more conflicting spreadsheets. Cost knowledge is captured, not locked in individuals.

VISION & NEXT STEPS

Intelligent, Instant, Integrated — What Comes Next

CostVision is live today. Here is what comes next — and how to get started.

 Current Capabilities — Live Today

- 21 commodity should-cost models — fully parametric, engineering-grade
- AI Agent — describe a part in plain English, AI builds the cost model
- AI CAD Analysis — STEP / photo → geometry → cost in minutes
- Self-audit — re-checks every estimate, corrects known errors, geometry stays truth
- Learns from actuals — bulk-import quotes, per-segment calibration + drift watch
- Universal machine-sizing — right press / tonnage picked across all commodities
- Assembly BOM Rollup — multi-part, multi-commodity assemblies
- Learning Curve (Wright's Law) + Supplier Quote Comparison + A/B/C scenarios
- DFM / DFA Intelligence — bolt-on AI layer, 1-10 scores
- 20 global regions · 10 currencies with live auto-switching
- 6-sheet Excel + professional PDF report with part photo
- Cloud sync, team sharing & secure JWT authentication
- 1,005 automated tests · CAD engine ships in a CI-verified container

Phase 2 — H2 2026

PLM/ERP integration connectors (Teamcenter, SAP)
Cost-target waterfall dashboard for programme teams
Multi-supplier quotation comparison matrix
Harden the Fly.io cloud deploy to multi-tenant

Phase 3 — 2027

Auto-tune calibration from a live ERP actuals feed
Autonomous sourcing RFQ generation from should-cost output
Executive portfolio analytics dashboard
Wider CAD coverage — assemblies and IGES edge cases

Phase 4 — 2027+

Digital Twin cost model (live update as the design changes)
Generative AI component redesign suggestions
Supplier risk scoring + supply-chain resilience index
Mobile app for shop-floor and supplier visits

 Pilot Rollout — Next Steps

- Pilot scope: 2-3 commodity families, one programme team
- Duration: 4-6 weeks with real parts vs supplier quotes
- Success metric: ≥15% supplier price reduction identified per part
- Contact: Avinash Bhosale — Cost Engineering & Digital Innovation

"Every engineer should have instant access to accurate, defensible should-cost intelligence — at concept stage, across every commodity, in every region." — Avinash Bhosale